
(a)
AdamW
Muon
step
35
75
150
300
450
750
1500
2.69
d0
2.09
d0
2.03
d1
1.55
1.56
1.56
1.32
1.32
1.33
(b)
JS (bits)
step
AdamW vs Muon
AdamW, other data order
Muon, other data order
(c)
JS (bits)
training loss
(d)
validation loss
step
AdamW
Muon